WORKED SOLUTION

STEP III 2015, Question 12

Probability and Statistics • Probability Generating Functions

STEP III 2015, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q12

The key idea

A remainder on division by m only cares about the exponent modulo m, so in the generating function of a remainder we may replace x^m by 1 wherever it appears. Adding independent throws multiplies generating functions, and then we reduce the powers of x again. With this rule, the whole calculation becomes simple polynomial algebra in which $x \cdot y$ and y^2 collapse to multiples of y.

Solution

Throughout, let $X_1, X_2, \ldots$ be the outcomes of the throws, which are independent and each uniform on $\{1, \ldots, 6\}$.

The reduction rule. Let m be a positive integer. If U and V are independent and take values in $\{0, 1, \ldots, m-1\}$, the pgf of $U+V$ is $\mathrm{G}_U(x)\mathrm{G}_V(x)$, a polynomial in which the term x^k carries the probability that $U+V=k$. The remainder of $U+V$ on division by m equals j exactly when $U+V$ is j or $j+m$. So we get the pgf of that remainder by replacing every x^k with x^{k-m} for $k \geq m$; in other words, by setting $x^m = 1$. Since the remainder of S_n is the remainder of (remainder of S_{n-1}) + (remainder of X_n), and these two are independent, we get the pgf of each remainder from the one before by multiplying and then setting $x^m = 1$. We write $\equiv$ for "equal once $x^m = 1$ is used".

Part (i)

Throws of $1, \ldots, 5$ leave remainders $1, \ldots, 5$, and a 6 leaves remainder 0. Each happens with probability $\frac{1}{6}$, so

$$\mathrm{G}(x) = \tfrac{1}{6}\left(1 + x + x^2 + x^3 + x^4 + x^5\right) = \tfrac{1}{6}z, \qquad \text{where } z = 1 + x + \cdots + x^5.$$

With $x^6 = 1$, multiplying z by x just moves each term round one place: $xz = x + x^2 + \cdots + x^5 + x^6 \equiv z$. Hence $x^k z \equiv z$ for every k, and

$$z^2 = \sum_{k=0}^{5} x^k z \equiv 6z.$$

So the pgf of R_2 is

$$\tfrac{1}{36} z^2 \equiv \tfrac{1}{36} \cdot 6z = \tfrac{1}{6} z = \mathrm{G}(x),$$

as required.

The same step then works at every stage: if R_{n-1} has pgf $\mathrm{G}(x)$, then R_n has pgf $\mathrm{G}(x)^2 \equiv \mathrm{G}(x)$. By induction, every R_n has pgf $\mathrm{G}(x) = \frac{1}{6} z$. The probability that $6 \mid S_n$ is $\mathrm{P}(R_n = 0)$, the constant term of $\mathrm{G}(x)$:

$$\mathrm{P}(S_n \text{ divisible by } 6) = \tfrac{1}{6}.$$

Part (ii)

Now $m = 5$. Throws of $1, 2, 3, 4$ give remainders $1, 2, 3, 4$; a 5 gives 0; a 6 gives 1. So remainder 1 has probability $\frac{2}{6}$ and the others $\frac{1}{6}$ each:

$$\mathrm{G}_1(x) = \tfrac{1}{6}\left(1 + 2x + x^2 + x^3 + x^4\right) = \tfrac{1}{6}(x + y), \qquad y = 1 + x + x^2 + x^3 + x^4.$$

Exactly as in part (i), with $x^5 = 1$ we have $xy \equiv y$, so $x^k y \equiv y$ for all k and

$$y^2 = \sum_{k=0}^{4} x^k y \equiv 5y.$$

Therefore

$$\begin{aligned} \mathrm{G}_2(x) &\equiv \tfrac{1}{36}(x + y)^2 = \tfrac{1}{36}\left(x^2 + 2xy + y^2\right) \\ &\equiv \tfrac{1}{36}\left(x^2 + 2y + 5y\right) = \tfrac{1}{36}\left(x^2 + 7y\right), \end{aligned}$$

which is the required result. (As a sanity check, putting $x = 1$ gives $\frac{1}{36}(1 + 35) = 1$.)

The general pgf. By the reduction rule, $\mathrm{G}_n(x) \equiv \mathrm{G}_1(x)^n = \frac{1}{6^n}(x + y)^n$. Expand by the binomial theorem:

$$(x + y)^n = x^n + \sum_{k=1}^{n} \binom{n}{k} x^{n-k} y^k.$$

For $k \geq 1$, using $y^k \equiv 5^{k-1} y$ (induction on $y^2 \equiv 5y$) and then $x^{n-k} y \equiv y$,

$$x^{n-k} y^k \equiv 5^{k-1} y.$$

So

$$(x + y)^n \equiv x^n + y \sum_{k=1}^{n} \binom{n}{k} 5^{k-1} = x^n + \frac{y}{5}\left[(1 + 5)^n - 1\right] = x^n + \frac{6^n - 1}{5}\, y.$$

Ask yourself Where did $(6^n - 1)/5$ come from?

The sum $\sum_{k=0}^{n} \binom{n}{k} 5^k$ is $(1+5)^n = 6^n$ by the binomial theorem; we are missing the $k = 0$ term (which is 1) and have an extra factor $\frac{1}{5}$. Hence

$$G_n(x) = \frac{1}{6^n}\left(x^r + \frac{6^n - 1}{5}\, y\right),$$

where r is the remainder when n is divided by 5 (since $x^n \equiv x^r$). So every remainder has probability $\dfrac{6^n - 1}{5 \cdot 6^n}$, except remainder r, which gets an extra $\frac{1}{6^n}$.

Divisibility by 5. We need $P(T_n = 0)$, the constant term of $G_n(x)$. The constant term of y is 1.

- If $5 \nmid n$, then $r \neq 0$ and x^r contributes nothing to the constant term, so

$$P(S_n \text{ divisible by } 5) = \frac{6^n - 1}{5 \cdot 6^n} = \frac{1}{5}\left(1 - \frac{1}{6^n}\right),$$

  as required.
- If $5 \mid n$, then $r = 0$ and $x^r = 1$ adds $\frac{1}{6^n}$:

$$P(S_n \text{ divisible by } 5) = \frac{6^n - 1}{5 \cdot 6^n} + \frac{1}{6^n} = \frac{6^n + 4}{5 \cdot 6^n} = \frac{1}{5}\left(1 + \frac{4}{6^n}\right).$$

Checks

- $n = 1$: the formula gives $\frac{1}{5} \cdot \frac{5}{6} = \frac{1}{6}$, and indeed only a throw of 5 is divisible by 5.
- $n = 2$: the formula gives $\frac{1}{5} \cdot \frac{35}{36} = \frac{7}{36}$. Directly, sums of 5 (four ways) and 10 (three ways) give $\frac{7}{36}$, matching the constant term of $\frac{1}{36}(x^2 + 7y)$.
- $n = 5$: the answer is $\frac{6^5+4}{5 \cdot 6^5} = \frac{389}{1944} \approx 0.2001$, just above $\frac{1}{5}$. Full enumeration of all 6^n outcomes by computer for $n \leq 8$ confirms the whole pgf $G_n(x)$ and both cases.
- As $n \to \infty$ both answers tend to $\frac{1}{5}$: the remainder becomes uniform.

Common slips

- Forgetting that a 6 leaves remainder 1 on division by 5, so that G_1 has coefficient $\frac{2}{6}$ for x.
- Reducing x^n to x^0 for every n: the term x^n reduces to x^r with r the remainder of n, which is exactly why the case $5 \mid n$ is different.
- Writing $G_n(x) = G_1(x)^n$ with "=" and then reading off coefficients without reducing modulo $x^5 = 1$.

Beyond STEP

Setting $x^m = 1$ amounts to working in the ring of polynomials modulo $x^m - 1$, the group algebra of the cyclic group of order m. An equivalent tool is the roots-of-unity filter: $\mathrm{P}(m \mid S_n) = \frac{1}{m}\sum_{\omega^m=1} \mathrm{G}_S(\omega)$, which is the discrete Fourier transform at work. That viewpoint also explains why sums of random steps on a cycle become uniform, the start of the theory of Markov chains and random walks on groups. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and random walks at the right level.

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